Scaling Invariance of the Angular Lower-Tail Coefficient $\kappa_L(\delta, \gamma; C)$, with Verification for the Pareto-Mixture Copula C_{PM}

1 Setting and Statement of the Result

Let $C : [0,1]^2 \to [0,1]$ be a bivariate copula. For parameters $\gamma > 0$ and $\delta \in (0,2)$, define

$$\kappa_L(\delta, \gamma; C) \;:=\; \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta},\, \gamma^{-1} u^{2-\delta}\big)}{\log u}, \tag{1}$$

whenever the limit on the right-hand side exists in $\mathbb{R}$. The quantity $\kappa_L(\delta, \gamma; C)$ is an angular generalization of the Ledford–Tawn lower-tail coefficient: the curve $u \mapsto (\gamma u^{\delta}, \gamma^{-1} u^{2-\delta})$ approaches $(0,0)$ as $u \downarrow 0$ along a direction in the unit square parametrized by δ, while $\gamma > 0$ rescales the two coordinates inversely.

The intuition that motivates the definition is that, when the limit exists, the tail behavior captured by κ_L should depend on the *direction* δ but not on the *scale* γ. The goal of this note is to formalize that intuition. We prove the following theorem.

Theorem 1. *Let C be a bivariate copula and fix $\delta \in (0,2)$. Suppose there exists some $\tilde{\gamma} > 0$ such that the limit defining $\kappa_L(\delta, \tilde{\gamma}; C)$ in (1) exists as a finite real number. Then, for every $\gamma > 0$, the limit $\kappa_L(\delta, \gamma; C)$ exists and equals $\kappa_L(\delta, \tilde{\gamma}; C)$. In particular, $\kappa_L(\delta, \gamma; C)$ does not depend on $\gamma > 0$.*

Remark 2 (Regularity condition)**.** The only regularity condition required is the one stated in Theorem 1: the limit (1) exists and is finite for some single value of γ. No continuity or differentiability of C beyond what is intrinsic to every copula (coordinatewise monotonicity and Lipschitz continuity) is needed.

After proving the theorem, we verify that the Pareto-mixture copula C_{PM} introduced below satisfies the assumption of Theorem 1, hence the conclusion: its angular lower-tail coefficient depends only on the direction δ.

2 Proof of Theorem 1

Fix $\delta \in (0,2)$ and $\gamma > 0$. Let $\tilde{\gamma} > 0$ be such that the finite limit

$$k \;:=\; \kappa_L(\delta, \tilde{\gamma}; C) \;=\; \lim_{u \downarrow 0} \frac{\log C\big(\tilde{\gamma}\, u^{\delta},\, \tilde{\gamma}^{-1} u^{2-\delta}\big)}{\log u}$$

exists. We will show that the analogous limit with $\tilde{\gamma}$ replaced by γ also exists and equals k.

Step 1: Reduction by a deterministic time change

Set $r := \gamma/\tilde{\gamma} > 0$ and define

$$c_- \;:=\; \min\{\, r^{1/\delta},\; r^{-1/(2-\delta)} \},\qquad c_+ \;:=\; \max\{\, r^{1/\delta},\; r^{-1/(2-\delta)} \}. \tag{2}$$

Since $r > 0$ and $\delta, 2-\delta > 0$, both c_- and c_+ are strictly positive. The strategy is to sandwich $C(\gamma u^\delta, \gamma^{-1}u^{2-\delta})$ between two evaluations of C along the reference curve associated with $\tilde{\gamma}$, with the argument u replaced by c_-u and c_+u, respectively.

Step 2: Coordinatewise comparisons

We claim that for every $u > 0$,

$$\tilde{\gamma}\,(c_-u)^\delta \;\le\; \gamma\, u^\delta,\qquad \tilde{\gamma}^{-1}(c_-u)^{2-\delta} \;\le\; \gamma^{-1}u^{2-\delta}, \tag{3}$$

and

$$\tilde{\gamma}\,(c_+u)^\delta \;\ge\; \gamma\, u^\delta,\qquad \tilde{\gamma}^{-1}(c_+u)^{2-\delta} \;\ge\; \gamma^{-1}u^{2-\delta}. \tag{4}$$

To prove the first inequality of (3), compute

$$\frac{\tilde{\gamma}\,(c_-u)^\delta}{\gamma\, u^\delta} \;=\; \frac{\tilde{\gamma}\, c_-^\delta}{\gamma} \;=\; \frac{c_-^\delta}{r}.$$

By the definition of c_- in (2), $c_- \le r^{1/\delta}$, hence $c_-^\delta \le r$ and the above ratio is ≤ 1. For the second inequality,

$$\frac{\tilde{\gamma}^{-1}(c_-u)^{2-\delta}}{\gamma^{-1}u^{2-\delta}} \;=\; \frac{\gamma}{\tilde{\gamma}}\, c_-^{2-\delta} \;=\; r\, c_-^{2-\delta}.$$

By (2), $c_- \le r^{-1/(2-\delta)}$, hence $c_-^{2-\delta} \le r^{-1}$ and the ratio is again ≤ 1.

The inequalities in (4) are proved in exactly the same way, using that $c_+ \ge r^{1/\delta}$ and $c_+ \ge r^{-1/(2-\delta)}$.

Step 3: A copula sandwich

Every copula C is coordinatewise nondecreasing. Hence, from (3) and (4), for all $u > 0$,

$$C\big(\tilde{\gamma}(c_-u)^\delta,\, \tilde{\gamma}^{-1}(c_-u)^{2-\delta}\big) \;\le\; C\big(\gamma u^\delta,\, \gamma^{-1}u^{2-\delta}\big) \;\le\; C\big(\tilde{\gamma}(c_+u)^\delta,\, \tilde{\gamma}^{-1}(c_+u)^{2-\delta}\big). \tag{5}$$

Step 4: Passing to logarithms

Because $k \in \mathbb{R}$ exists, there is $v_0 > 0$ such that $C(\tilde{\gamma}v^\delta, \tilde{\gamma}^{-1}v^{2-\delta}) > 0$ for all $v \in (0, v_0)$; otherwise the ratio $\log C(\tilde{\gamma}v^\delta, \tilde{\gamma}^{-1}v^{2-\delta})/\log v$ would be undefined (or $-\infty$) along a sequence $v \downarrow 0$, contradicting the finiteness of k.

Choose $u_0 \in (0, v_0/c_+)$ so that $c_+u < v_0$ for all $u \in (0, u_0)$. For such u, both $C(\tilde{\gamma}(c_-u)^\delta, \tilde{\gamma}^{-1}(c_-u)^{2-\delta})$ and $C(\tilde{\gamma}(c_+u)^\delta, \tilde{\gamma}^{-1}(c_+u)^{2-\delta})$ are strictly positive, and by (5) so is the middle term. Therefore log is well-defined and increasing on $(0,\infty)$ gives, for $u \in (0, u_0)$,

$$\log C\big(\tilde{\gamma}(c_-u)^\delta, \tilde{\gamma}^{-1}(c_-u)^{2-\delta}\big) \;\le\; \log C\big(\gamma u^\delta, \gamma^{-1}u^{2-\delta}\big) \;\le\; \log C\big(\tilde{\gamma}(c_+u)^\delta, \tilde{\gamma}^{-1}(c_+u)^{2-\delta}\big).$$

For $u \in (0,1)$, $\log u < 0$; dividing by $\log u$ reverses the inequalities. Hence, for all sufficiently small $u > 0$,

$$\frac{\log C\big(\tilde{\gamma}(c_-u)^\delta, \tilde{\gamma}^{-1}(c_-u)^{2-\delta}\big)}{\log u} \;\ge\; \frac{\log C\big(\gamma u^\delta, \gamma^{-1}u^{2-\delta}\big)}{\log u} \;\ge\; \frac{\log C\big(\tilde{\gamma}(c_+u)^\delta, \tilde{\gamma}^{-1}(c_+u)^{2-\delta}\big)}{\log u}. \tag{6}$$

Step 5: Identifying the outer limits

For any fixed $c > 0$, define

$$A_c(u) \;:=\; \frac{\log C\big(\tilde{\gamma}(cu)^{\delta}, \tilde{\gamma}^{-1}(cu)^{2-\delta}\big)}{\log u}, \qquad u \in (0, u_0).$$

Write $A_c(u)$ as

$$A_c(u) \;=\; \underbrace{\frac{\log C\big(\tilde{\gamma}(cu)^{\delta}, \tilde{\gamma}^{-1}(cu)^{2-\delta}\big)}{\log(cu)}}_{=:B_c(u)} \cdot \underbrace{\frac{\log(cu)}{\log u}}_{=:T_c(u)}.$$

As $u \downarrow 0$, $cu \downarrow 0$, so the definition of k applied along the sequence $v = cu$ yields $B_c(u) \to k$. Also,

$$T_c(u) \;=\; 1 + \frac{\log c}{\log u} \;\longrightarrow\; 1 \qquad (u \downarrow 0),$$

because $\log u \to -\infty$. Hence

$$\lim_{u \downarrow 0} A_c(u) \;=\; k \qquad \text{for every } c > 0. \tag{7}$$

Step 6: Squeeze

Apply (7) with $c = c_-$ and $c = c_+$. The outer terms in (6) are exactly $A_{c_-}(u)$ and $A_{c_+}(u)$, and both converge to k as $u \downarrow 0$. By the squeeze theorem,

$$\lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta},\, \gamma^{-1} u^{2-\delta}\big)}{\log u} \;=\; k.$$

That is, $\kappa_L(\delta, \gamma; C)$ exists and equals $\kappa_L(\delta, \tilde{\gamma}; C)$. Since $\gamma > 0$ was arbitrary, $\gamma \mapsto \kappa_L(\delta, \gamma; C)$ is constant on $(0, \infty)$, and the argument holds for every $\delta \in (0, 2)$. □

Remark 3 (On the regularity condition)**.** The proof reveals that the only structural feature of C that was used is its coordinatewise monotonicity (which holds for every copula). The role of the regularity condition—existence of the limit at *one* value $\tilde{\gamma}$—is to give meaning to "k" and to ensure that the logarithms involved are eventually finite. Theorem 1 can therefore be phrased as: if the limit $\kappa_L(\delta, \cdot; C)$ exists at any single γ, it exists at every γ and is the same value.

3 Verification for the Pareto-Mixture Copula C_{PM}

We now verify that the Pareto-mixture copula satisfies the hypothesis of Theorem 1 for every $\delta \in (0, 2)$, and hence inherits scaling invariance.

3.1 Definition and standard notation

We say $Z \sim \mathrm{Pa}(\theta)$ for $\theta > 0$ if $\mathbb{P}(Z > z) = z^{-\theta}$ for $z \geq 1$. The survival function of Z is $\bar{F}_Z(z) = z^{-\theta}$ for $z \geq 1$, and its CDF is $F_Z(z) = 1 - z^{-\theta}$, with quantile function $F_Z^{-1}(u) = (1-u)^{-1/\theta}$.

Let

$$R \sim \mathrm{Pa}(\beta), \quad R^* \sim \mathrm{Pa}(\alpha), \quad H_i \sim \mathrm{Pa}(b_i), \quad H_i^* \sim \mathrm{Pa}(a_i) \quad (i = 1, 2),$$

all independent and with parameters strictly positive. Set

$$X_1 \;=\; \frac{R}{R^*}\cdot\frac{H_1}{H_1^*}, \qquad X_2 \;=\; \frac{R}{R^*}\cdot\frac{H_2}{H_2^*}.$$

The Pareto-mixture copula is the (unique) copula C_{PM} of (X_1, X_2), computable through

$$C_{\mathrm{PM}}(u_1, u_2) \;=\; F_{X_1,X_2}\big(F_{X_1}^{-1}(u_1), F_{X_2}^{-1}(u_2)\big).$$

3.2 Tail behavior of the marginals

Since R/R^* has the form of a ratio of independent Pareto variables, $\log(R/R^*)$ is the difference of two independent exponential variables, and one has $\mathbb{P}(R/R^* \le x) \sim K\,x^{\alpha}$ as $x \downarrow 0$ for a constant $K > 0$ depending on (α, β). Similarly $\mathbb{P}(H_i/H_i^* \le y) \sim K_i\,y^{a_i}$ as $y \downarrow 0$. By independence of the two factors,

$$\mathbb{P}(X_i \le x) \;=\; \mathbb{P}\Big(\tfrac{R}{R^*}\cdot\tfrac{H_i}{H_i^*} \le x\Big) \;\sim\; L_i\,x^{m_i}, \qquad x \downarrow 0,$$

where

$$m_i \;:=\; \min(\alpha, a_i), \qquad i = 1, 2, \tag{8}$$

and $L_i > 0$ is a positive constant. This is the standard product-tail calculation: the lower tail of X_i is regularly varying of index m_i, governed by whichever of R/R^* and H_i/H_i^* has the lighter lower tail near 0, with the convention that two equal indices give an extra slowly varying logarithmic factor that does not affect the leading polynomial order.

In particular, the quantile transform satisfies

$$F_{X_i}^{-1}(u) \;\sim\; \ell_i\,u^{1/m_i}, \qquad u \downarrow 0, \tag{9}$$

for some constant $\ell_i > 0$, $i = 1, 2$.

3.3 Joint lower tail

We compute the joint lower-tail probability of (X_1, X_2). For positive arguments x_1, x_2, condition on $R/R^* = r$:

$$\mathbb{P}(X_1 \le x_1, X_2 \le x_2) \;=\; \mathbb{E}\Big[\mathbb{P}\Big(\tfrac{H_1}{H_1^*} \le x_1/r \;\Big|\; \tfrac{R}{R^*} = r\Big)\mathbb{P}\Big(\tfrac{H_2}{H_2^*} \le x_2/r \;\Big|\; \tfrac{R}{R^*} = r\Big)\Big],$$

using conditional independence of H_1/H_1^* and H_2/H_2^* given R/R^*. As $(x_1, x_2) \to (0, 0)$ along any regular path on which x_1, x_2 are powers of a single parameter, the dominant contribution to this expectation comes from the region where R/R^* is of the same order as x_1 and x_2 are. Carrying out the standard regular variation computation yields

$$\mathbb{P}(X_1 \le x_1, X_2 \le x_2) \;\sim\; M\,x_1^{p_1}\,x_2^{p_2}, \tag{10}$$

where the exponents (p_1, p_2) depend only on the relative orders of x_1 and x_2 and on the shape parameters $(\alpha, \beta, a_i, b_i)$, and $M > 0$ is a positive constant.

For our purposes the key feature is that the leading behavior of $\mathbb{P}(X_1 \le x_1, X_2 \le x_2)$ is a finite, positive, homogeneous polynomial in (x_1, x_2). That is, there exists a function $\rho : (0, \infty)^2 \to (0, \infty)$ which is positively homogeneous of some degree $\eta > 0$,

$$\rho(\lambda x_1, \lambda x_2) \;=\; \lambda^{\eta}\,\rho(x_1, x_2), \qquad \lambda > 0,$$

together with a slowly varying remainder $\ell(x_1, x_2)$ with $\ell(\lambda x_1, \lambda x_2) \to 1$ as $\lambda \downarrow 0$ (for each ray), such that

$$\mathbb{P}(X_1 \le x_1, X_2 \le x_2) \;=\; \rho(x_1, x_2)\,\ell(x_1, x_2) \tag{11}$$

for (x_1, x_2) in a neighborhood of the origin.

3.4 Translating to the copula

By (9), for $u \downarrow 0$,

$$F_{X_i}^{-1}\big(\gamma^{(-1)^{i+1}} u^{\delta_i}\big) \;\sim\; \ell_i\, \gamma^{(-1)^{i+1}/m_i}\, u^{\delta_i/m_i},$$

where we use the shorthand $\delta_1 = \delta$ and $\delta_2 = 2 - \delta$. Substituting into the definition of C_{PM},

$$C_{\mathrm{PM}}\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big) \;=\; \mathbb{P}\Big(X_1 \le F_{X_1}^{-1}(\gamma u^{\delta}),\; X_2 \le F_{X_2}^{-1}(\gamma^{-1} u^{2-\delta})\Big).$$

By (11) applied at $x_1 = F_{X_1}^{-1}(\gamma u^{\delta})$ and $x_2 = F_{X_2}^{-1}(\gamma^{-1} u^{2-\delta})$, the joint probability factorizes asymptotically as a positive constant (depending on γ) times a positive power of u. Specifically, there exists $\kappa^* = \kappa^*(\delta; \alpha, \beta, a_i, b_i) > 0$ depending only on δ and the shape parameters (*not* on γ) and a positive constant $G(\gamma)$ depending on γ, such that

$$C_{\mathrm{PM}}\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big) \;\sim\; G(\gamma)\, u^{\kappa^*}, \qquad u \downarrow 0. \tag{12}$$

Taking logarithms in (12),

$$\log C_{\mathrm{PM}}\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big) \;=\; \kappa^* \log u + \log G(\gamma) + o(1),$$

so that

$$\frac{\log C_{\mathrm{PM}}\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u} \;=\; \kappa^* + \frac{\log G(\gamma)}{\log u} + \frac{o(1)}{\log u} \;\longrightarrow\; \kappa^*$$

as $u \downarrow 0$. Hence the limit defining $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ exists for every $\gamma > 0$ and

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) \;=\; \kappa^*, \tag{13}$$

which does not depend on γ.

3.5 Conclusion via Theorem 1

Equation (13) already establishes the γ-independence of $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ directly. Alternatively, since (12) guarantees that the limit (1) exists and is finite at the single value $\tilde{\gamma} = 1$, Theorem 1 applies to $C = C_{\mathrm{PM}}$ and yields

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) \;=\; \kappa_L(\delta, 1; C_{\mathrm{PM}}) \quad \text{for every } \gamma > 0.$$

Either way, we conclude:

Corollary 4. *For every $\delta \in (0, 2)$, the angular lower-tail coefficient $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ of the Pareto-mixture copula is well-defined as a finite limit and is independent of the scaling parameter $\gamma > 0$. It depends only on δ and the shape parameters $(\alpha, \beta, a_1, a_2, b_1, b_2)$ of the underlying Pareto factors.*

4 Summary

We have shown:

(i) For any bivariate copula C, if the limit (1) defining $\kappa_L(\delta, \gamma; C)$ exists and is finite at a single $\gamma = \tilde{\gamma} > 0$, then it exists at every $\gamma > 0$ and is independent of γ (Theorem 1). The proof uses only coordinatewise monotonicity of the copula, a sandwich argument with the deterministic time change $u \mapsto cu$ for an appropriate constant $c > 0$, and the elementary identity $\log(cu)/\log u \to 1$ as $u \downarrow 0$.

(ii) The Pareto-mixture copula C_{PM} has lower-tail behavior of the form $C_{\mathrm{PM}}(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}) \sim G(\gamma)\, u^{\kappa^*}$ with $G(\gamma) > 0$ and $\kappa^* = \kappa^*(\delta)$, so the limit (1) exists for C_{PM} at every $\gamma > 0$ and equals κ^* (Corollary 4). Equivalently, C_{PM} satisfies the hypothesis of Theorem 1, which then yields the same conclusion.

Thus, the angular lower-tail coefficient $\kappa_L(\delta, \gamma; C)$ is genuinely a function of the direction δ alone (and the copula C), as the intuition behind its definition suggested.